

PROGRAM ASSESSMENT DATA AY2026

UNITED STATES MILITARY ACADEMY
DEPARTMENT OF CHEMISTRY AND LIFE SCIENCE
CHEMICAL ENGINEERING PROGRAM
August 10, 2026

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Level of Achievement of Student Outcome 1:

On completion of the chemical engineering program, our graduates will be able to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Fundamentals of Engineering Examination, once/yr.
3. End-of-Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

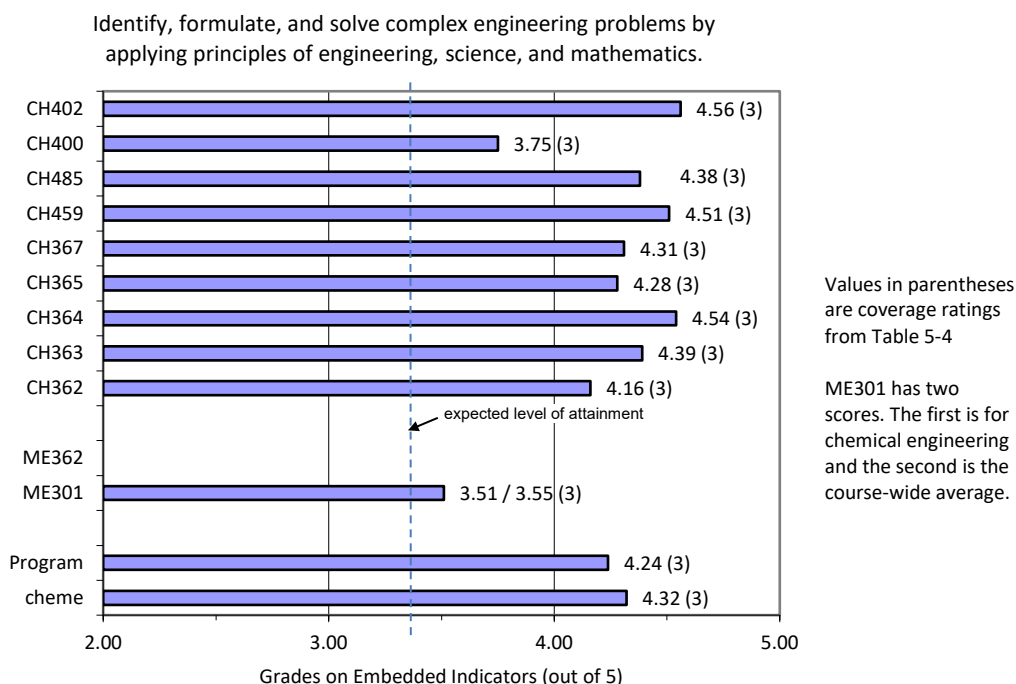


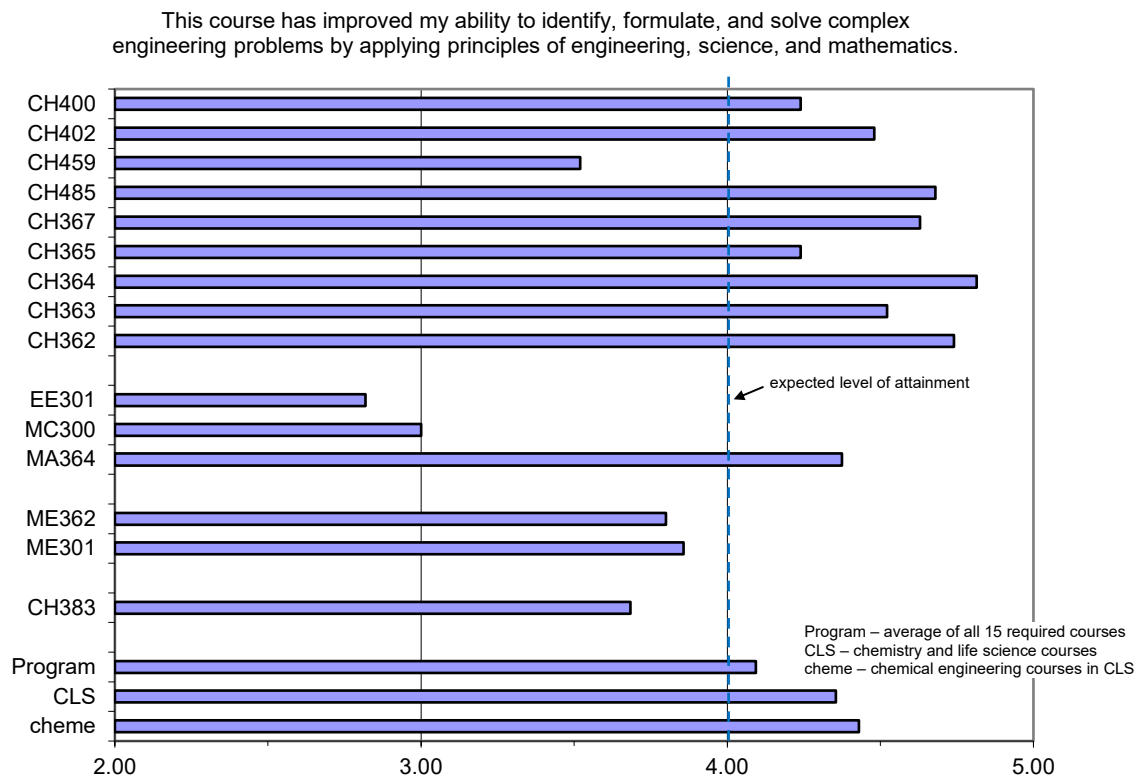
Figure 1-1. Coursework Embedded Indicator results for Student Outcome 1.

2. Fundamentals of Engineering Examination (FEE). According to the 2026 report from NCEES, 19 out of 25 cadets or 76% in the Class of 2026 took and passed the FEE. The national average passing in 2026 was 67.9% (483 out of 711), and this is our expected level of attainment. In the previous five years, the pass rates were 63.3% in 2025, 78.9% in 2024, 76.9% in 2023, 91.3% in 2022, and 85.0% in 2021. Our running average over those previous five years is $79.1\% \pm 10.5\%$ versus $71.8\% \pm 1.7\%$ for the national average (uncertainties are standard deviations).

Note on FEE results: We assess our program against the national average, which is our expected level of attainment, and we are **above** the national average pass rate for last year and the running average.

3. End of Semester Student Surveys

Figure 1-2. End-of-Semester Student Survey responses for Student Outcome 1.



4. Chemical Engineering Program Exit Survey. This survey is issued to Firsties at the end of their last semester. Cadets were asked whether they agree with the statement “The program has prepared me to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.” 25 out of 25 cadets completed the survey. Of these, 22 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 16/25 replying that they strongly agreed (score = 5/5) and 6/25 that they agreed (score = 4/5). The mean is $4.40/5.00 \pm 1.04$ for the 25 cadets participating in the survey. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 2:

On completion of the chemical engineering program, our graduates will be able to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Course Grades in CH402 Chemical Engineering Process Design, once/yr.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

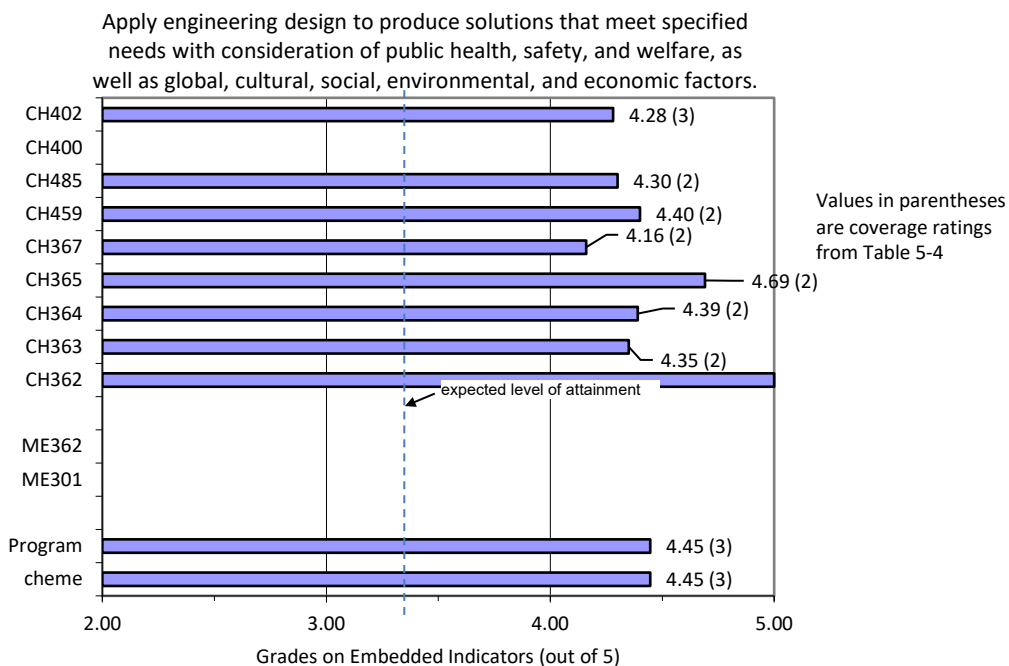


Figure 2-1. Coursework Embedded Indicator results for Student Outcome 2.

2. End of Semester Student Surveys

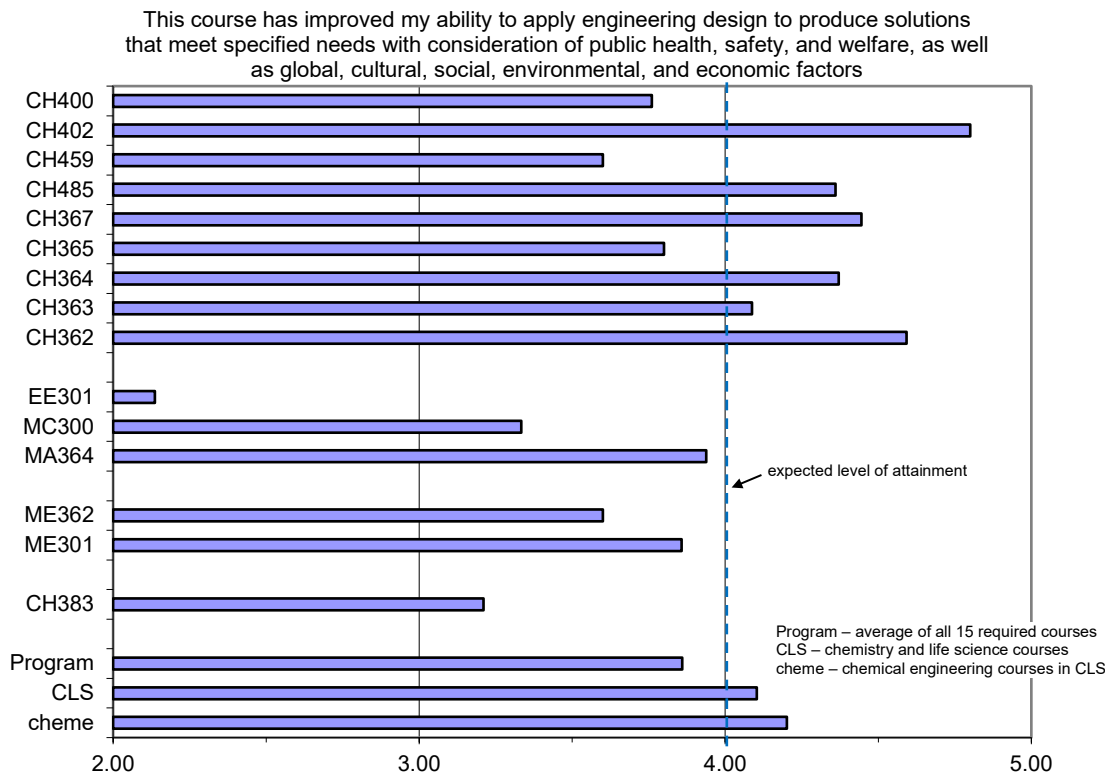


Figure 2-2. End-of-Semester Student Survey responses for Student Outcome 2.

3. The average course grade in CH402 Chemical Engineering Process Design was 3.20 ± 0.48 ($n=25$) in AY26, compared to 3.34 ± 0.70 ($n=29$) in AY25, 3.65 ± 0.46 ($n=19$) in AY24, 3.44 ± 0.55 ($n=13$) in AY23, 3.33 ± 0.50 ($n=23$) in AY22, and 3.46 ± 0.68 ($n=20$) in AY21. The 5-year running average for the previous five years is 3.44, and this is our expected level of attainment. This year's score was *below* the 5-year running average, primarily due to weaker performance on the written design reports.
4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether or not they agree with the statement "The program has prepared me to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors." 25 out of 25 cadets completed the survey. Of these, 22 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 13/25 replying that they strongly agreed (score = 5/5) and 9/25 that they agreed (score = 4/5). The mean is $4.32/5.00 \pm 0.90$ for the 25 cadets. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 3:

On completion of the chemical engineering program, our graduates will be able to communicate effectively with a range of audiences.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Course Grades in CH459 Unit Operations Laboratory, once/yr.
4. Chemical Engineering Program Exit Survey, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

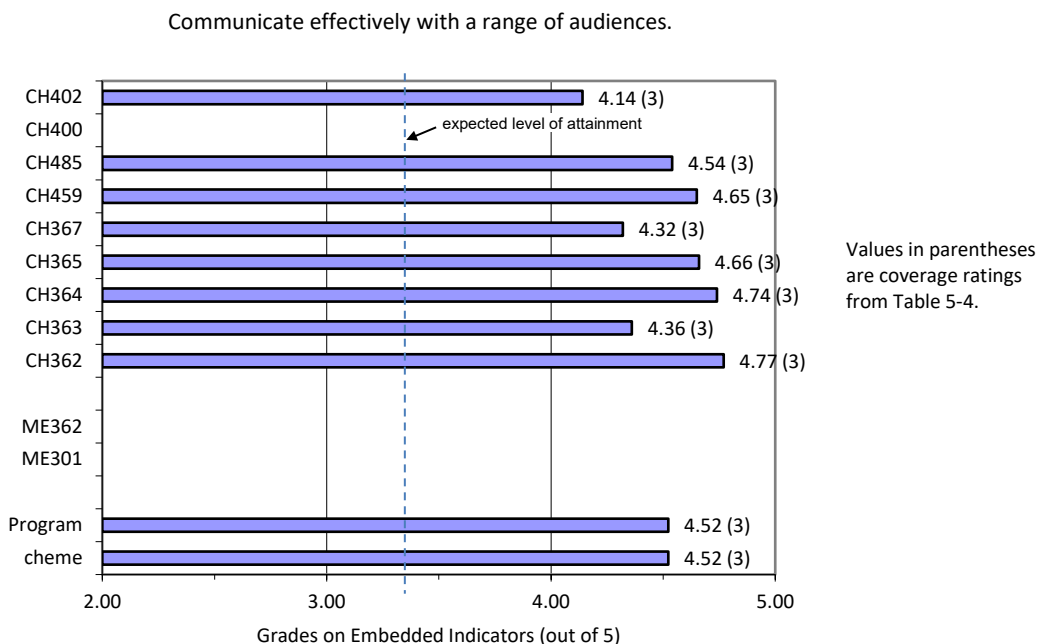


Figure 3-1. Coursework Embedded Indicator results for Student Outcome 3.

2. End of Semester Student Surveys

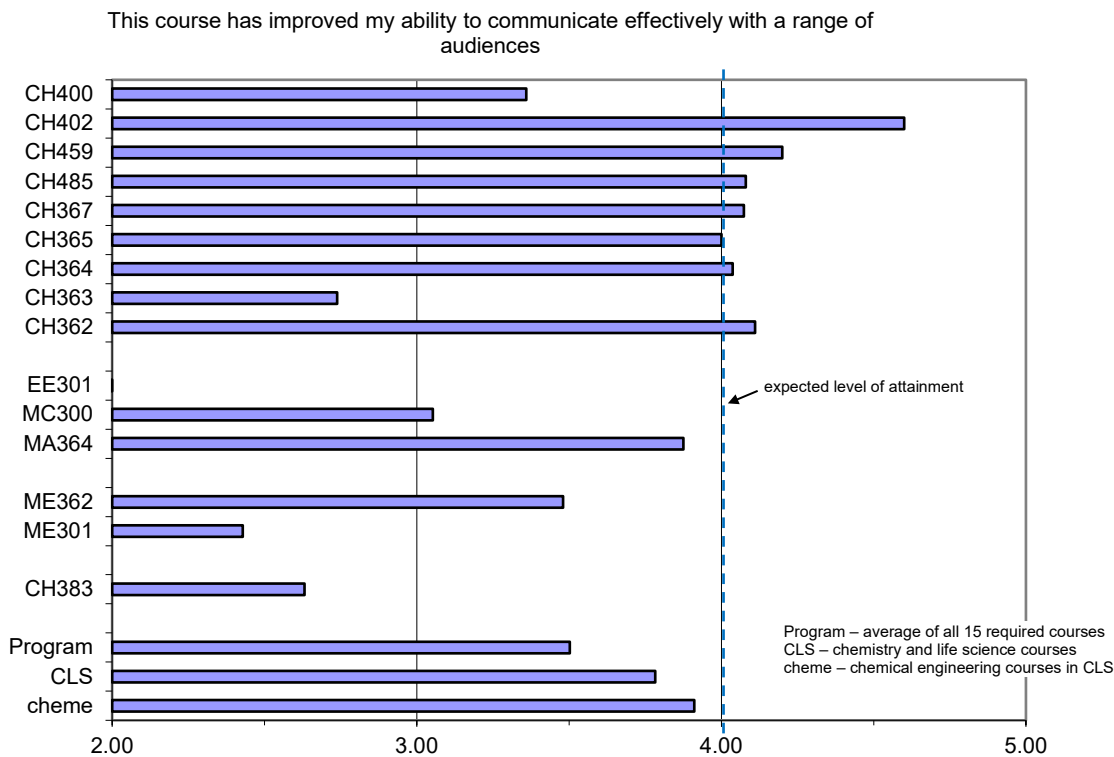


Figure 3-2. End-of-Semester Student Survey responses for Student Outcome 3.

3. The average course grade in CH459 Chemical Engineering Laboratory was 3.60 ± 0.54 ($n=25$) in AY26, compared to 3.52 ± 0.59 ($n=29$) in AY25, 3.70 ± 0.46 ($n=19$) in AY24, 3.41 ± 0.43 ($n=13$) in AY23, 3.80 ± 0.42 ($n=23$) in AY22, and 3.67 ± 0.53 ($n=18$) in AY21. The 5-year running average is 3.62, and this is our expected level of attainment. This year's score was *essentially the same as* the 5-year running average.
4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether they agree with the statement "The program has prepared me to communicate effectively with a range of audiences." 25 out of 25 cadets completed the survey. Of these, 22 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 13/25 replying that they strongly agreed (score = 5/5) and 9/25 that they agreed (score = 4/5). The mean is $4.24/5.00 \pm 1.13$ for the 25 cadets. The expected level of attainment on this survey is 4.00/5.00.

Level of Achievement of Student Outcome 4:

On completion of the chemical engineering program, our graduates will be able to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Fundamentals of Engineering Examination Performance Index, once/yr.
3. End of Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.
5. Completion of Cadet Character Education Program, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

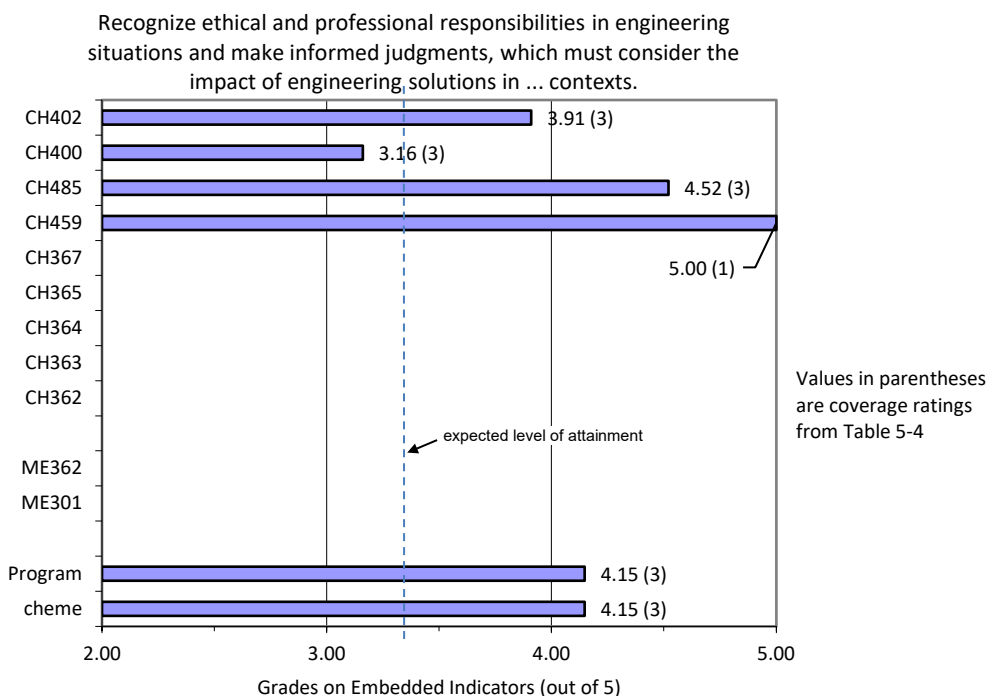


Figure 4-1. Coursework Embedded Indicator results for Student Outcome 4.

2. Fundamentals of Engineering Examination Performance, Self-Study Table 4-2.

Subject	Outcome	Questions	USMA	National (Expected level of attainment)
Ethics and Professional Practice	4	3	11.6	11.3 ± 4.4
Economics	4	4	11.0	9.6 ± 3.6

Note on FEE results: We assess our program against the national average, which is our expected level of attainment, and we are **above** the national average pass rate in both of these topics.

3. End of Semester Student Surveys

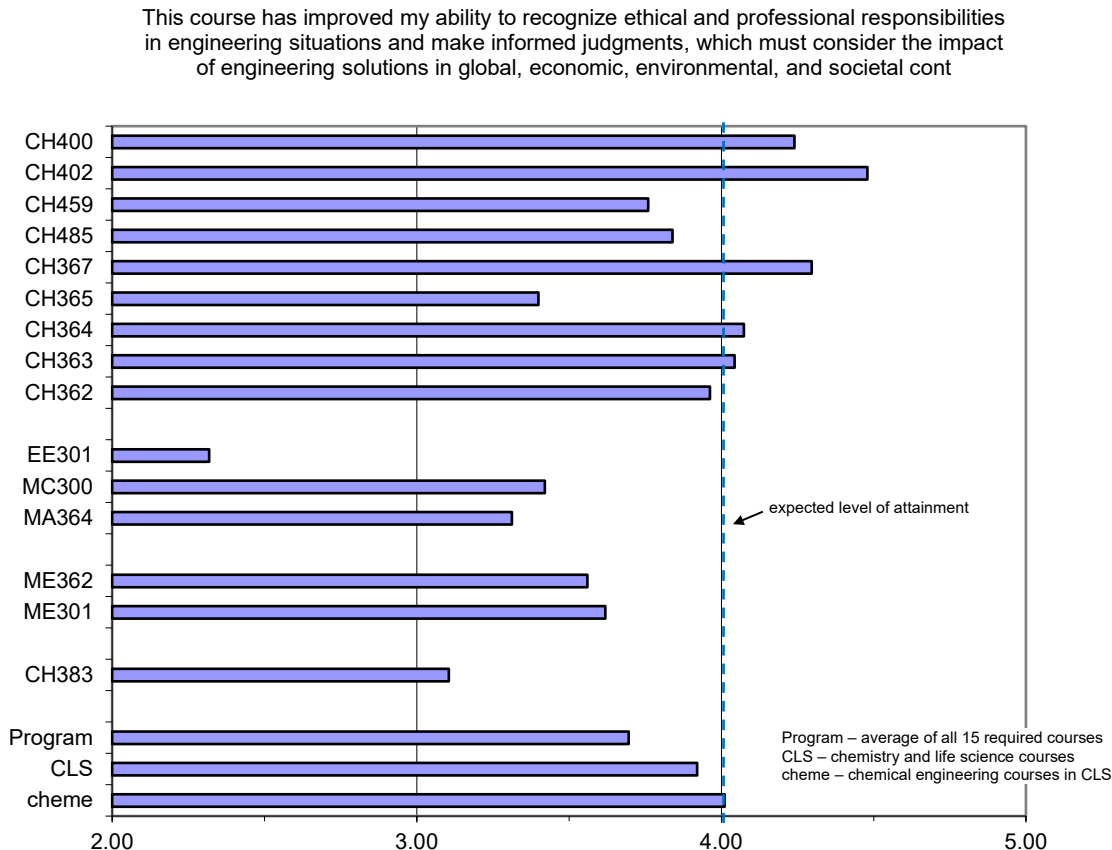


Figure 4-2. End-of-Semester Student Survey responses for Student Outcome 4.

4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to Firsties at the end of their last semester. They were asked whether they agree with the statement “The program has prepared me to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.” 25 out of 25 cadets completed the survey. Of these, 19 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 9/25 replying that they strongly agreed (score = 5/5) and 10/25 that they agreed (score = 4/5). The mean is 4.00/5.00 $\pm$ 1.00 for the 25 cadets. The expected level of attainment is 4.00/5.00.
5. Training in honor and ethics takes place in the Cadet Character Education Program (CCEP) during the academic year and summer terms. The program is overseen by the Commandant of Cadets through the Simon Center for the Professional Military Ethic. CCEP customizes instruction to each of the four year-groups of cadets, who interact with faculty volunteers who share their perspectives and experience in the Armed Forces, with industry, and at other civilian institutions. All 23 chemical engineering cadets successfully completed the 4-year CCEP program.

Level of Achievement of Student Outcome 5:

On completion of the chemical engineering program, our graduates will be able to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End of Semester Student Surveys, once/semester.
3. Chemical Engineering Program Exit Survey, once/yr.
4. Teamwork Skills Rubric, twice/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

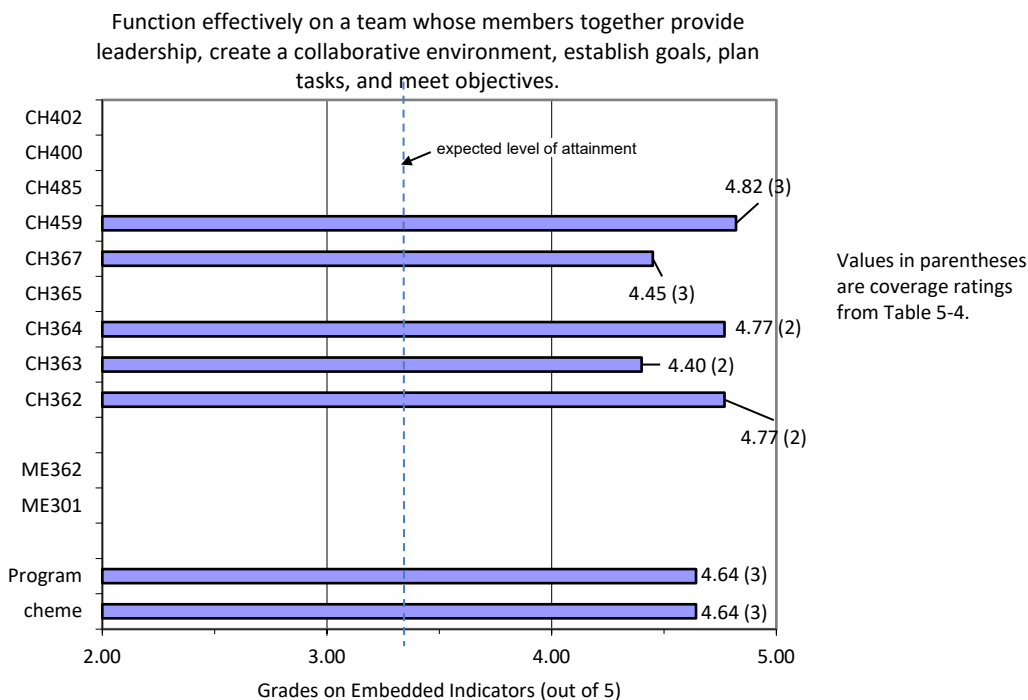


Figure 5-1. Coursework Embedded Indicator results for Student Outcome 5.

2. End of Semester Student Surveys

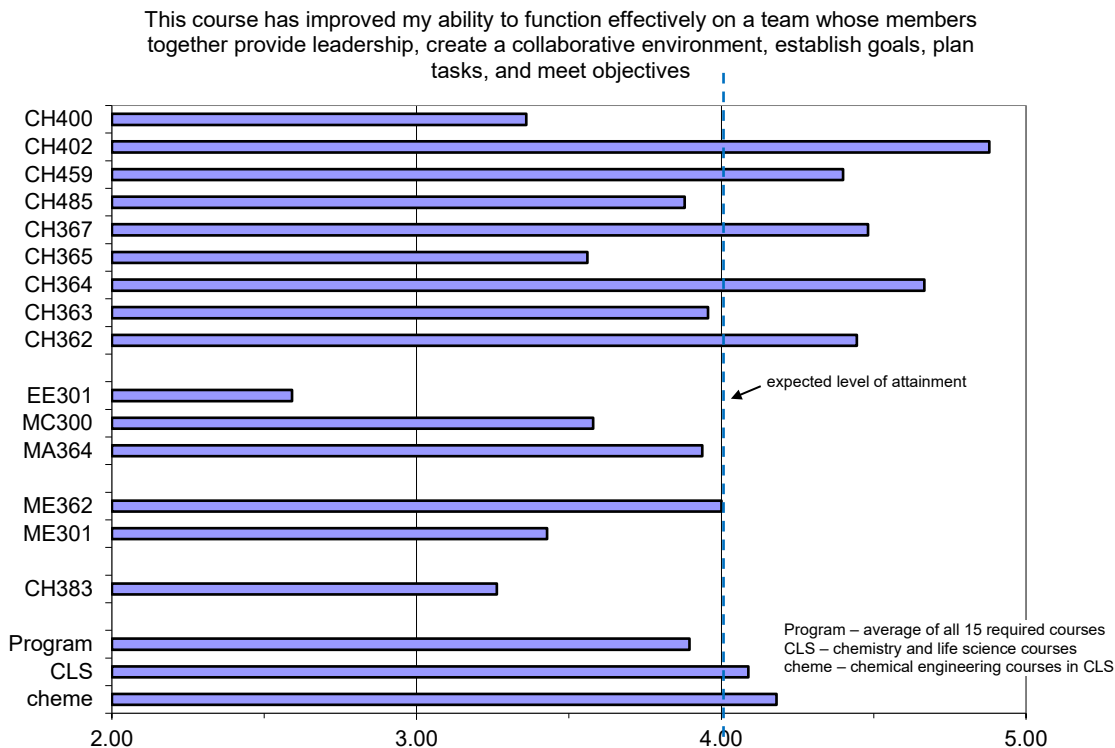


Figure 5-2. End-of-Semester Student Survey responses for Student Outcome 5.

3. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether or not they agree with the statement “The program has prepared me to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.” 25 out of 25 cadets completed the survey. Of these, 22 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 14/25 replying that they strongly agreed (score = 5/5) and 8/25 that they agreed (score = 4/5). The mean is $4.28/5.00 \pm 1.14$ for the 25 cadets. The expected level of attainment on this survey is 4.00/5.00.
4. Teamwork Skills Rubric. The rubric is used to assess performance in four skills associated with the ability to function on effectively on a team, namely technical competence, communication, organization, and teamwork. The rubrics are completed by the cadets after each laboratory exercise in CH459. The results were averaged over each round-robin and reported below using the actual rubric for formatting. The superscript “a” next to the averages are for team assessment of the group leader and “b” designates group leader assessment of the team. The expected level of attainment is 4.0. The color shading indicates the relationship of the averages to the expected level of attainment. Red shading indicates when improvement is needed.

Your Name: Dr. Simuck F. Yuk				Person Assessed: Cadets in CH459					
Your Position: CH459 CD				Major of Person Assessed: Chemical Engineering					
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5	N/A
Technical Competence	Some misunderstandings of the technical content.			Demonstrated knowledge of the technical content.			Exceptional knowledge of technical content.	4.8 ^a ±.36 ^c 4.7 ^b ±.47 ^c	
Communication	Lacked sensitivity and/or did not provide specific suggestions for improvement.			Effectively communicated important points.			Exceptional ability to explain important points. Very effectively communicated ideas for improvement.	4.7 ^a ±.42 ^c 4.6 ^b ±.45 ^c	
Organization	Was not prepared or did not give sufficient time to prepare.			Demonstrated effective organization during class.			Was exceptionally efficient, timely and responsive throughout the entire process.	4.8 ^a ±.45 ^c 4.6 ^b ±.48 ^c	
Teamwork	Demonstrated limited ability to see other perspectives or find common ground.			Worked collaboratively with team members to reach consensus.			Exceptional ability to help group find common ground or resolve conflict in order to ultimately reach consensus.	4.8 ^a ±.40 ^c 4.6 ^b ±.44 ^c	
Are the cadets capable of functioning on multidisciplinary teams? Yes	Comments: Footnote “a” designates the average of all Team Assessments of Leader (TAL) scores, while “b” designates average of all Group Leader Assessment (GLAT) scores. Footnote “c” designates standard deviations.						Assignment used for assessment: AY26, Round Robin 1		

Your Name: Dr. Simuck F. Yuk				Person Assessed: Cadets in CH459					
Your Position: CH459 CD				Major of Person Assessed: Chemical Engineering					
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5	N/A
Technical Competence	Some misunderstandings of the technical content.			Demonstrated knowledge of the technical content.			Exceptional knowledge of technical content.	4.6 ^a ±.71 ^c 4.7 ^b ±.60 ^c	
Communication	Lacked sensitivity and/or did not provide specific suggestions for improvement.			Effectively communicated important points.			Exceptional ability to explain important points. Very effectively communicated ideas for improvement.	4.6 ^a ±.70 ^c 4.7 ^b ±.49 ^c	
Organization	Was not prepared or did not give sufficient time to prepare.			Demonstrated effective organization during class.			Was exceptionally efficient, timely and responsive throughout the entire process.	4.6 ^a ±.64 ^c 4.7 ^b ±.63 ^c	
Teamwork	Demonstrated limited ability to see other perspectives or find common ground.			Worked collaboratively with team members to reach consensus.			Exceptional ability to help group find common ground or resolve conflict in order to ultimately reach consensus.	4.6 ^a ±.68 ^c 4.7 ^b ±.54 ^c	
Are the cadets capable of functioning on multidisciplinary teams? Yes	Comments: Footnote “a” designates the average of all Team Assessments of Leader (TAL) scores, while “b” designates average of all Group Leader Assessment (GLAT) scores. Footnote “c” designates standard deviations.						Assignment used for assessment: AY26, Round Robin 2		

Level of Achievement of Student Outcome 6:

On completion of the chemical engineering program, our graduates will be able to develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. End-of-Semester Student Surveys, once/semester.
3. Chemical Engineering Program Exit Survey, once/yr.
4. Course Grades in CH459 Unit Operations Laboratory, once/yr.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

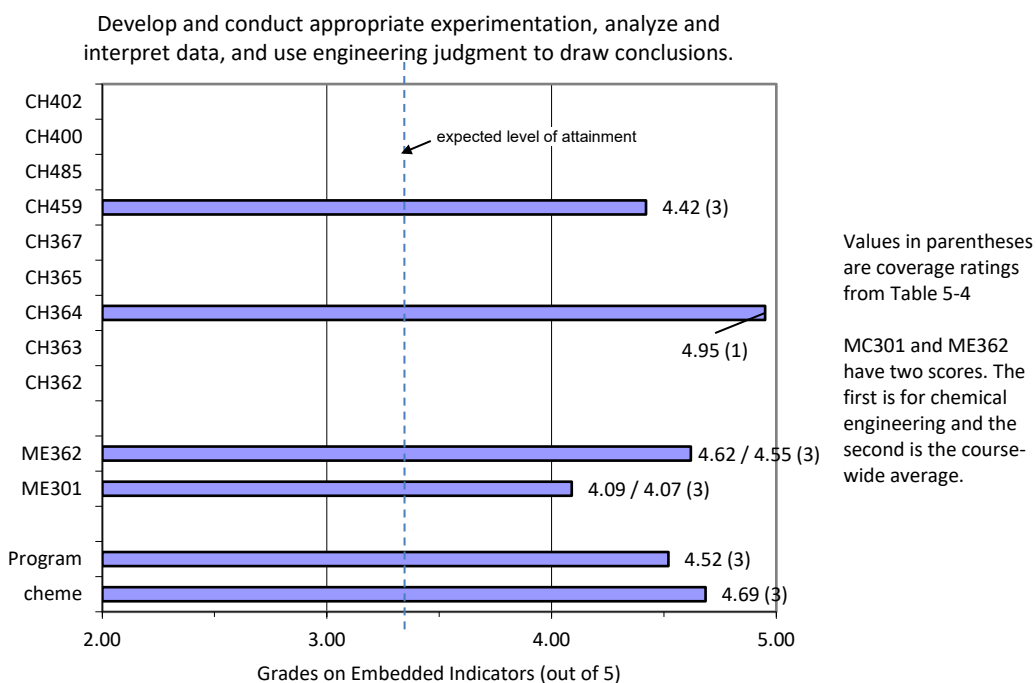


Figure 6-1. Coursework Embedded Indicator results for Student Outcome 6.

2. End of Semester Student Surveys

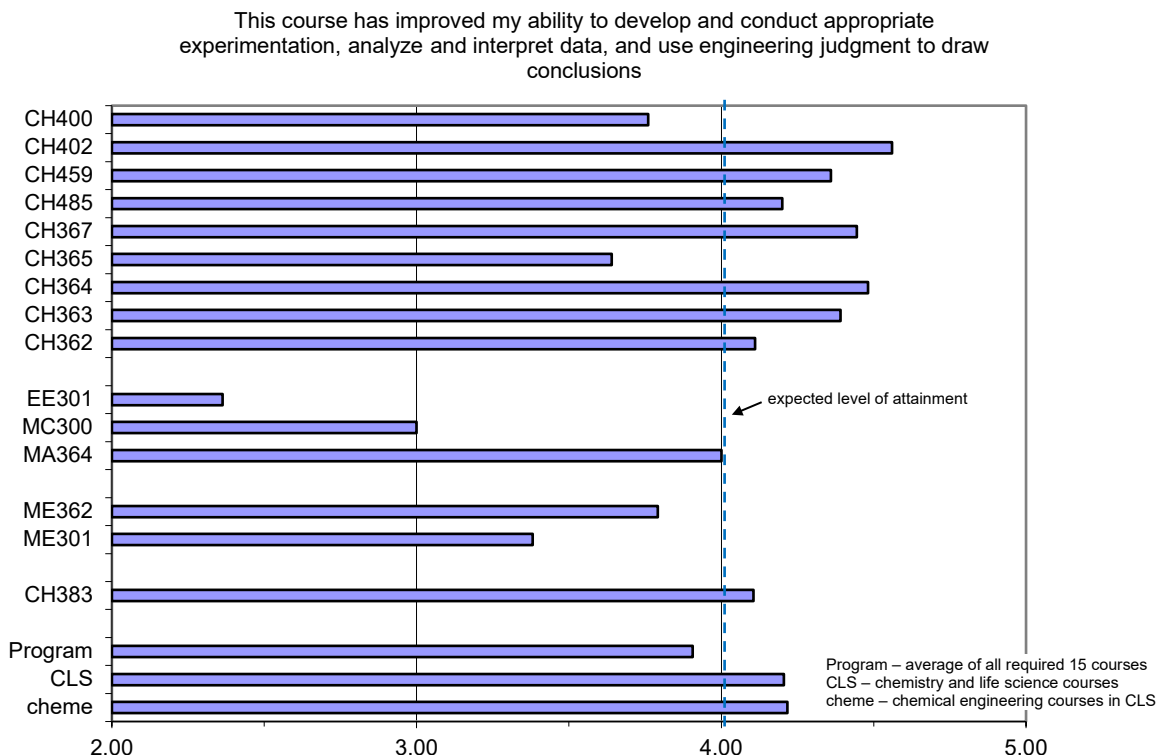


Figure 6-2. End-of-Semester Student Survey responses for Student Outcome 6.

3. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether or not they agree with the statement “The program has prepared me to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.” 25 out of 25 cadets completed the survey. Of these, 21 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 15/29 replying that they strongly agreed (score = 5/5) and 6/25 that they agreed (score = 4/5). The mean is $4.32/5.00 \pm 1.07$ for the 25 cadets. The expected level of attainment on this survey is 4.00/5.00.
4. The average course grade in CH459 Chemical Engineering Laboratory was 3.60 ± 0.54 (n=25) in AY26, compared to 3.52 ± 0.59 (n=29) in AY25, 3.70 ± 0.46 (n=19) in AY24, 3.41 ± 0.43 (n=13) in AY23, 3.80 ± 0.42 (n=23) in AY22, and 3.67 ± 0.53 (n=18) in AY21. The 5-year running average is 3.62, and this is our expected level of attainment. This year’s score was *essentially the same* as the 5-year running average.

Level of Achievement of Student Outcome 7:

On completion of the chemical engineering program, our graduates will be able to acquire and apply new knowledge as needed, using appropriate learning strategies.

Assessment Instruments and Frequency:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators, once/yr.
2. Percent of eligible students taking the Fundamentals of Engineering Examination (FEE), once/yr.
3. End of Semester Student Surveys, once/semester.
4. Chemical Engineering Program Exit Survey, once/yr.
5. Lifelong Learning Skills Rubric, twice per year.
6. Contemporary Issues Rubric, twice per year.

Assessment Results:

1. Chemical & Mechanical Engineering Coursework Embedded Indicators

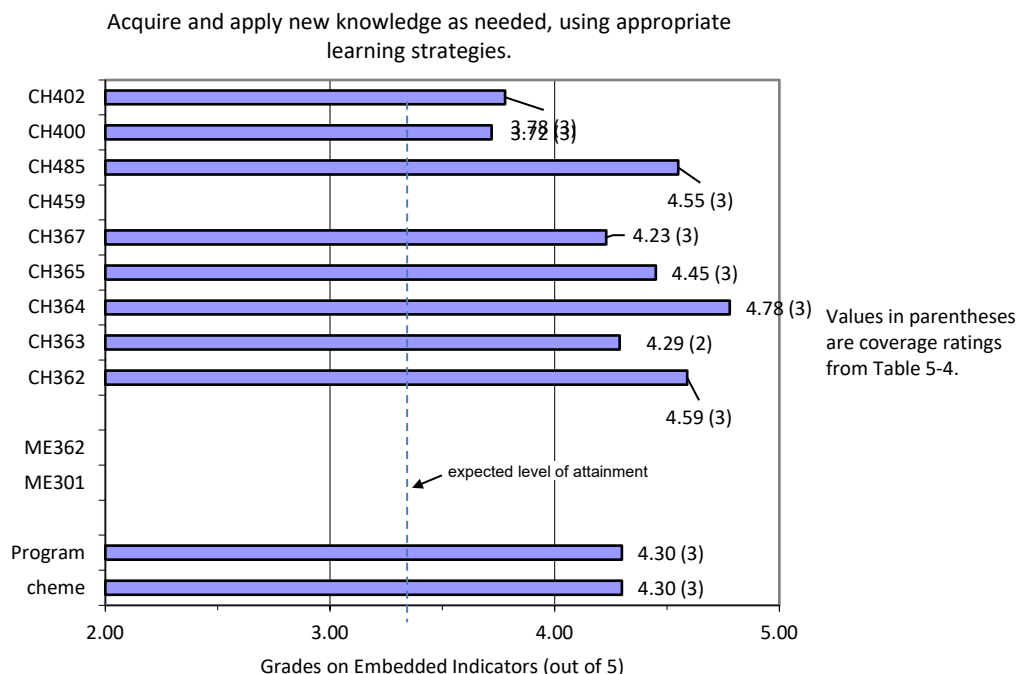


Figure 7-1. Coursework Embedded Indicator results for Student Outcome 7.

2. Percent of Eligible Cadets taking the Fundamentals of Engineering Examination (FEE). For the Class of 2026, 25 of 25 chemical engineering cadets (100% of eligible cadets) prepared for and took the FEE. For comparison, the participation rates in the previous five years were 29 of 29 in the class of 2025, 19 of 19 in 2024, 13 of 13 in 2023, 23 of 23 in 2022, and 20 of 20 in 2021 (100% of eligible cadets in all years). **Note: The participation rate is EXCELLENT.**

3. End of Semester Student Surveys

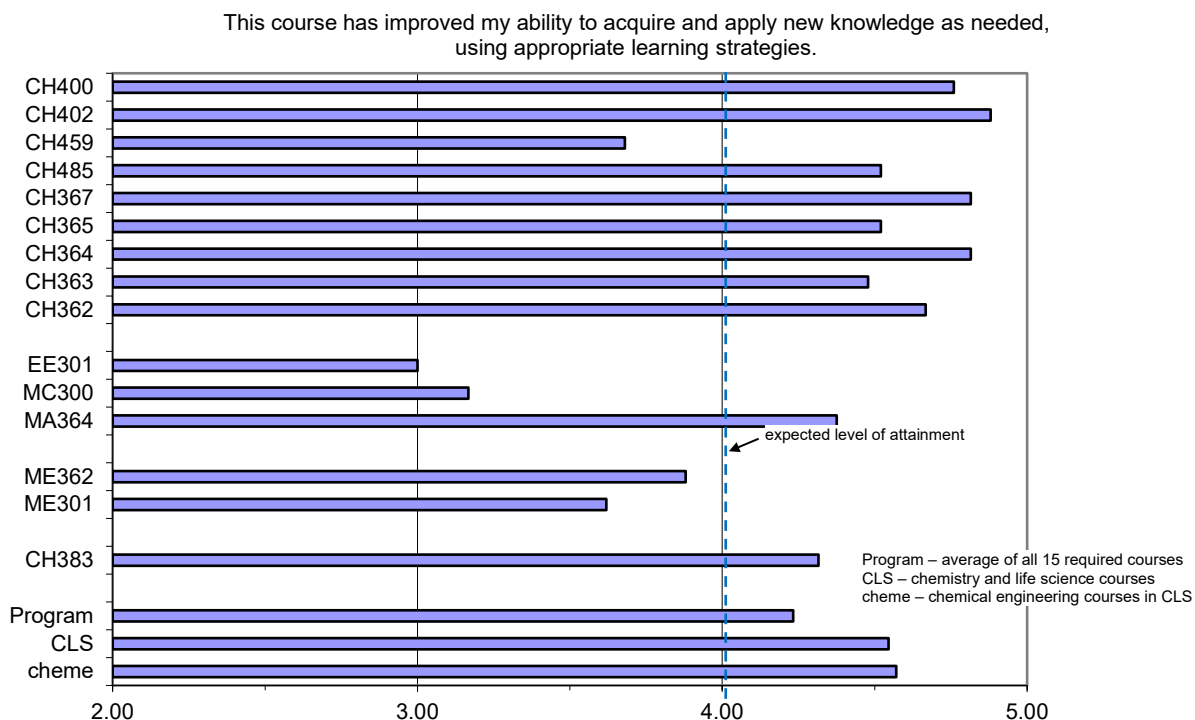


Figure 7-2. End-of-Semester Student Survey responses for Student Outcome 7.

4. Chemical Engineering Program Exit Survey. As stated earlier, this survey is given to the Firsties at the end of their last semester. They were asked whether they agree with the statement “The program has prepared me to acquire and apply new knowledge as needed, using appropriate learning strategies.” 25 out of 25 cadets completed the survey. Of these, 21 out of 25 cadets said that they either agreed or strongly agreed (score = 5/5 or 4/5), with 14/29 replying that they strongly agreed (score = 5/5) and 7/25 that they agreed (score = 4/5). The mean is $4.36/5.00 \pm 0.86$ for the 25 cadets. The expected level of attainment on this survey is 4.00/5.00.
5. Lifelong Learning Skills Rubric. This rubric, when used in tandem with the resume writing assignment in CH365, is designed to assess performance in four skills associated with the ability to acquire new knowledge, namely: rubric row 1 - engagement (in professional activities), rubric row 2 - recognition (of skills learned in the program), rubric row 3 – intellectual growth (recognition of new skills), and rubric row 4 – communication (in resume format). Resumes were written at the beginning of the semester (assignment 1.1), then revised at the end of the semester (assignment 2.1). One rubric is shown for each assignment below, along with cadet averages and standard deviations. The expected levels of attainment are color-coded, with red indicating a need for improvement, yellow indicating acceptable

performance, and green indicating that expectations are met or exceeded. Instructor comments are in the rubrics.

Instructor's Name: Biaglow				Cadet Assessed: Summary, all cadets in CH365				
Your Position: Instructor, CH365 (e.g., CD CH365)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Engagement Outcome 7	No evidence of pre-professional activities.			References to pre-professional activities are lacking or connections to chemical engineering are weak or implied.			Uses examples of pre-professional chemical engineering activities.	4.9 ± 0.4
Recognition Outcome 7	Skills learned in chemical engineering courses taken in previous semesters are not listed.			Skills are listed, but the skills are vaguely described, or connection to chemical engineering concepts is not clear.		4.4 ± 0.9	Identifies specific skills learned in chemical engineering courses.	
Intellectual Growth Outcome 7	Unable to identify new concepts learned this semester.			Changes are apparent in document, but connections to recent activities in chemical engineering are weak or implied.	3.2 ± 0.9		Addition of multiple skills acquired this semester.	
Communication Outcome 3	Resume lacks organization or cohesion. Numerous grammatical errors that may interfere with meaning. Target audience unclear.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write a basic resume, but document is uninteresting and flat.		4.4 ± 1.1	Demonstrates an ability to effectively communicate in the resume format. Clear, concise content. Resume is interesting.	
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) N	Comments: Cadet scores were somewhat low overall due to the low score in row three. Scores were low in this row because many cadets are unable to articulate new skills learned this semester.					Assignment used for assessment: (e.g., resume draft 1.1 in CH365) resume draft 1 in CH365		
	Final Grade: 4.2 ± 1.1 / 5.0							

Instructor's Name: Biaglow				Cadet Assessed: Summary, all cadets in CH365				
Your Position: Instructor, CH365 (e.g., CD CH365)				Cadet Major: Chemical Engineering (e.g., Chem. Eng.)				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Engagement Outcome 7	No evidence of pre-professional activities.			References to pre-professional activities are lacking or connections to chemical engineering are weak or implied.			Uses examples of pre-professional chemical engineering activities.	4.7 ± 0.4
Recognition Outcome 7	Skills learned in chemical engineering courses taken in previous semesters are not listed.			Skills are listed, but the skills are vaguely described, or connection to chemical engineering concepts is not clear.			Identifies specific skills learned in chemical engineering courses.	4.7 ± 0.5
Intellectual Growth Outcome 7	Unable to identify new concepts learned this semester.			Changes are apparent in document, but connections to recent activities in chemical engineering are weak or implied.		4.4 ± 0.8	Addition of multiple skills acquired this semester.	
Communication Outcome 3	Resume lacks organization or cohesion. Numerous grammatical errors that may interfere with meaning. Target audience unclear.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write a basic resume, but document is uninteresting and flat.			Demonstrates an ability to effectively communicate in the resume format. Clear, concise content. Resume is interesting.	4.8 ± 0.3
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y	Comments: Overall cadet scores were driven up compared to draft 1 because cadets became more familiar with the expectations, particularly in row three, which increased by more than a full point. Overall, I think the exercise was a success. Final Grade: 4.6 ± 0.5 / 5.0						Assignment used for assessment: (e.g., resume draft 1.1 in CH365) resume draft 2 in CH365	

There is normally improvement in scores in draft 2 as cadets incorporate instructor comments to improve and polish their documents. Version 2, though, requires articulation of new skills acquired during the semester. Scores can go down if cadets do not show new activities and skills learned during the current semester, and in some individual cases the scores did drop, and intellectual growth (row 3) did increase. Overall and on average, the program scores are solidly in the green in all four categories by the end of the semester.

6. Contemporary Issues Rubric (follows on page 18). The contemporary issues rubric is designed to assess performance in four skills associated with the ability to acquire new knowledge, namely: rubric row 1 - acquiring knowledge (through development and use of references), rubric row 2 - technical competence of the cadet (as demonstrated with writing skills), rubric row 3 – impact of contemporary issues (ability to link contemporary issues to relevant chemical engineering concepts), and rubric row 4 - grammar and structure. Each of these skills, taken independently, are associated with written communication (Student Outcome 3). However, the development and blending of these skills into a coherent and well-crafted essay are a measure of acquisition of new knowledge and are thus used by our program to assess Student Outcome 7. Admittedly, this is not a complete measure of a student's ability to acquire new knowledge, only that ability as expressed in an essay. Therefore, we combine this exercise with the other embedded indicators in this section of the report.

Operationally, the contemporary issues rubric is completed by the instructor to award grades to cadets for the writing assignments in this course. In this case, LTC Cowart used one writing assignment in CH485, completed in two drafts. The draft and final submissions were graded against the rubric, with the total average score and average scores per outcome shown in the screenshots above. In AY26 (this year), the average for the draft was 4.1 ± 0.9 and the average on the final was 4.8 ± 0.6 , so cadets achieved a better average on the final than on the draft submission. Scores are also very close to the previous year, 4.1 ± 0.9 and 4.8 ± 0.6 in AY25 in the draft and final reports, respectively. We conclude that cadets achieved each of the graded rubric rows with proficiency. Instructor comments are included in the screenshots.

Instructor's Name: LTC Cowart				Cadet Assessed: All Cadets in CH485 (AY26-1)				
Your Position: CD, CH485				Cadet Major: Chemical Engineering				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Acquiring Knowledge <i>Outcome 7</i>	Cites references but they are not substantive or do not address assignment. Context is weak or implied. Presents generic information or lacks specificity.			Cites at least two references relevant to the assignment. References have been adequately studied and re-worded to inform the audience.		4.2 +/- 0.9	Uses multiple substantive examples or scholarly articles in an integrative fashion to fully inform the reader on the topic.	
Technical Competence <i>Outcomes 8</i>	Demonstrates poor or incomplete understanding of technical content.			Demonstrates some knowledge of the technical content, but explanation lacks adequate depth.		3.9 +/- 0.6	Demonstrates exceptional knowledge of technical content.	
Impact of Contemporary Issues <i>Outcome 4</i>	Does not have a premise or does not connect issues in essay to concepts in chemical engineering.			Makes connections to chemical engineering concepts, but the connections are weak or implied.		4.1 +/- 1.1	Makes very clear connections between the contemporary issues and chemical engineering concepts.	
Communication <i>Outcome 3</i>	Writing lacks organization or cohesion. Numerous grammatical errors or errors interfere with meaning.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write an essay but lacks cohesion or completeness.		4.3 +/- 0.9	Demonstrates an ability to effectively communicate in the essay format. Fully supported, clear, concise information. Writing style was exceptionally clear and articulate.	
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y Comments: Draft essays largely addressed one or more contemporary issues, and were relatively complete drafts of a full essay. Significant course-wide lack of technical depth was noted and source material was inadequate. In-text citation of source material was often lacking and several papers were not well organized. Assignment used for assessment: (e.g., draft 1 writing assignment 1 in CH485) CH485 Writing DRAFT Final Grade: 4.1 +/- 0.9								

Instructor's Name: LTC Cowart				Cadet Assessed: All Cadets in CH485 (AY26-1)				
Your Position: CD, CH485				Cadet Major: Chemical Engineering				
	1 – Needs Improvement	1	2	3 – Meets Expectations	3	4	5 – Exceeds Expectations	5
Acquiring Knowledge <i>Outcome 7</i>	Cites references but they are not substantive or do not address assignment. Context is weak or implied. Presents generic information or lacks specificity.			Cites at least two references relevant to the assignment. References have been adequately studied and re-worded to inform the audience.			Uses multiple substantive examples or scholarly articles in an integrative fashion to fully inform the reader on the topic.	4.8 +/- 0.5
Technical Competence <i>Outcomes 8</i>	Demonstrates poor or incomplete understanding of technical content.			Demonstrates some knowledge of the technical content, but explanation lacks adequate depth.			Demonstrates exceptional knowledge of technical content.	4.8 +/- 0.4
Impact of Contemporary Issues <i>Outcome 4</i>	Does not have a premise or does not connect issues in essay to concepts in chemical engineering.			Makes connections to chemical engineering concepts, but the connections are weak or implied.			Makes very clear connections between the contemporary issues and chemical engineering concepts.	4.8 +/- 0.4
Communication <i>Outcome 3</i>	Writing lacks organization or cohesion. Numerous grammatical errors or errors interfere with meaning.			Occasional grammar errors that do not impede meaning. Demonstrates ability to write an essay but lacks cohesion or completeness.			Demonstrates an ability to effectively communicate in the essay format. Fully supported, clear, concise information. Writing style was exceptionally clear and articulate.	4.8 +/- 0.4
Has this cadet demonstrated SO7 (acquiring knowledge)? (Y/N) Y Comments: Significant improvement from the draft version of the contemporary issues essay. Technical improvement, inclusion of figures/drawings/schematics, coupled with extensive discussion of contemporary issues regarding the chosen topic resulted in interesting, thoughtful papers. Assignment used for assessment: (e.g., draft 1 writing assignment 1 in CH485) CH485 Writing FINAL Final Grade: 4.8 +/- 0.4								

Level of Achievement of Student Outcome 8:

The program provides the graduate with a thorough grounding and working knowledge of the chemical sciences, including:

- (8.1) Chemistry.
- (8.2) Material and energy balances
- (8.3) Safety and environmental factors.
- (8.4) Thermodynamics of physical and chemical equilibria.
- (8.5) Heat, mass, and momentum transfer.
- (8.6) Chemical reaction engineering.
- (8.7) Continuous and staged separation operations.
- (8.8) Process dynamics and control.
- (8.9) Modern experimental and computing techniques.
- (8.10) Process design.

Assessment Instruments and Frequency:

1. Fundamentals of Engineering Examination, once/yr.
2. Average Course Grades for Chemical Engineering Students, once/yr.

Assessment Results:

1. Fundamentals of Engineering Examination, Self-Study Table 4-3. For the Class of 2026, the breakdown by topic is shown in comparison to the national averages.

Subject	Outcome	Questions	USMA ChE	National (expected level of attainment)
Chemistry & Biology	8.1	7	9.3	9.5 ± 2.6
Material & Energy Balances	8.2	10	9.4	9.2 ± 2.2
Safety, Health, & Environmental	8.3	5	8.7	9.5 ± 3.4
Thermodynamics	8.4	8	9.4	9.2 ± 2.0
Heat Transfer	8.5	8	9.9	9.8 ± 2.7
Fluid Mechanics/Dynamics	8.5	8	9.9	9.2 ± 2.3
Chemical Reaction Engineering	8.6	7	10.4	9.5 ± 2.5
Mass Transfer & Separations	8.7	8	9.7	9.5 ± 2.3
Process Control	8.8	4	10.4	9.8 ± 3.5
Computational Tools	8.9	0	---	---
Economics	8.10	4	11.0	9.6 ± 3.6
Process Design	8.10	7	10.3	10.2 ± 3.1

2. Course grades for the last six years are shown below, Table 4-4 (from Self Study).

Course ↓		Chemical Engineering Student Outcome 8								
		Advanced Chemistry	Mater. & Energy Bal.	Thermodynamics	Transport	Reaction Engineering	Separations	Dynamics & Control	Experiment & Compute	Process Design
		8.1	8.2	8.4	8.5	8.6	8.7	8.8	8.9	8.10
CH383	Organic Chemistry I	2.91								
CH365	Chem. Eng. Thermo.			3.95						
CH362	Mass & Energy Balances		3.04							
CH363	Separation Processes						3.40			
CH364	Chem. Reaction Eng.					3.21				
CH459	Chem. Eng. Laboratory								3.60	
CH485	Heat and Mass Transfer				3.45					
CH402	Chem. Eng. Process Des.									3.20
ME301	Thermodynamics			3.31						
ME362	Fluid Mechanics				3.51					
CE300	Fund. Eng. Mech. & Des.									
EE301	Intro. To Elec. Engineering									
CH367	Intr. Auto. Process Control							3.25		
Average Grade 2026		2.91	3.04	3.63	3.48	3.21	3.40	3.25	3.60	3.20
Average Grade 2025		2.97	3.36	3.60	3.45	3.25	3.66	3.36	3.52	3.34
Average Grade 2024		2.53	3.48	3.64	3.23	3.54	3.68	3.28	3.70	3.65
Average Grade 2023		2.64	3.17	3.35	3.21	3.38	3.59	3.51	3.41	3.44
Average Grade 2022		3.15	3.64	3.59	3.47	3.33	3.69	3.42	3.80	3.33
Average Grade 2021		2.78	3.54	3.33	3.23	3.31	3.37	3.35	3.67	3.46
Previous 5-year Running Average (Expected level of attainment)		2.81	3.44	3.48	3.34	3.36	3.60	3.38	3.62	3.44
Standard Deviation 2026 (N=25)		0.73	0.72	0.58	0.51	0.61	0.69	0.68	0.54	0.48

Table 4-1. Faculty Evaluation of Chemical Engineering Student Outcomes (N=13).

Chemical Engineering Student Outcomes	Faculty Evaluation
On completion of the chemical engineering program, our graduates are able to:	
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	4.85±0.38
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	4.54±0.66
3. Communicate effectively with a range of audiences.	4.62±0.77
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	4.54±0.66
5. Function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.	4.77±0.44
6. Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	4.77±0.60
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.	4.69±0.48
8. Understand the chemical engineering curriculum, including chemistry, material and energy balances, safety and environmental factors, thermodynamics of physical and chemical equilibria, heat, mass, and momentum transfer, chemical reaction engineering, continuous and staged separation processes, process dynamics and control, modern experimental and computing techniques, and process design.	4.92±0.28
5- Excellent; 4 – Very Good; 3 – Acceptable; 2 – Weak ; 1 – Poor	

Table 4-2. Advisory Board Evaluation of Chemical Engineering Student Outcomes.

Chemical Engineering Student Outcomes	Advisory Board's Evaluation
On completion of the chemical engineering program, our graduates are able to:	
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	
3. Communicate effectively with a range of audiences.	
4. Recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering decisions on global, economic, environmental, and societal contexts.	
5. Function effectively on teams to provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.	
6. Develop and conduct appropriate experimentation, analyze, and interpret data, and use engineering judgment to draw conclusions.	
7. Acquire and communicate knowledge as needed, using appropriate learning strategies.	
8. Understand the chemical engineering curriculum, including chemistry, materials, and energy balances, safety and environmental factors, thermodynamics of physical and chemical equilibria, heat, mass, and momentum transfer, chemical reaction engineering, continuous and staged separation processes, process dynamics and control, modern experimental and computing techniques, and process design.	
5- Excellent; 4 – Very Good; 3 – Acceptable; 2 – Weak ; 1 – Poor	

Advisory board scores are not yet available for AY26 and will be updated after the next meeting in the Spring of 2027.

Table 4-3. Summary of Chemical Engineering Student Outcomes Performance.

Chemical Engineering Student Outcomes	Program Director's Summary
On completion of the chemical engineering program, our graduates are able to:	
1. Identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.	
2. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.	
3. Communicate effectively with a range of audiences.	
4. Recognize ethical and professional responsibilities in engineering situations and make informed decisions, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.	
5. Function effectively on a team whose members together provide leadership, create a collaborative and supportive environment, establish goals, plan tasks, and meet objectives.	
6. Develop and conduct scientific experiments, analyze and interpret data, and use sound judgment to draw conclusions.	
7. Acquire and apply new knowledge as needed, using appropriate learning strategies.	
8. Understand and apply basic chemical engineering curriculum, including chemistry, physics, energy balances, safety and environmental factors, thermodynamics, kinetics of physical and chemical equilibria, heat, mass, and momentum transfer, chemical reaction engineering, continuous and staged separation processes, process dynamics and control, modern experimental and computing techniques, and process design.	
5- Excellent; 4 – Very Good; 3 – Acceptable; 2 – Weak ; 1 – Poor	